Doc2 — Software Requirements Specification: HabbitQuest

Nov 3, 2026 · @Rahul

1. Introduction

This specification defines what HabbitQuest v1 must do, as testable requirements traced to the business requirements (BR-xx) in Doc1.

Scope: the web app (parent area and kid mode) and the HabitQuest API (FastAPI and PostgreSQL), which also runs the AI helpers. Native apps are out of scope (see Doc1).

Definitions

Term	Meaning
Quest	A habit in a child's plan, done on its scheduled days: a catalog habit or one of the family's own
Routine time	A child's start time for a part of the day (morning, midday, after school, evening, bedtime); a habit's default time comes from its part of the day
Catalog habit	A pre-approved habit from the catalog (8 goals, 41 habits), for an age range, in tier 1, 2 or 3; a family can also make its own custom habits
Spark	Lifetime progress unit; an approved quest earns its tier (1-3), at most 6 a day; never removed
Points	Spendable currency earned on approval; spent in the reward shop
Level	One per world, set by lifetime Sparks: Starter Camp (0), Mosswood Forest (30), Sky Islands (90), Coral Cove (180), Crystal Caves (300), Star Harbour (450)
Kid mode	The child-facing screens; leaving needs the parent PIN when the family has one
Proposal / Recommendation	An AI-suggested plan change or new habit, applied only after a grown-up accepts it (with the PIN, if set)

Term	Meaning
Guardrails	Local and AI safety checks on text children read or write

References: Doc1 Product Vision and Business Requirements; Doc3 Solution Architecture; Doc4 Technical Design; repository rahulwarekar20/habbit-quest.

2. Overall description

HabbitQuest is a single-page web app (React) that calls the HabitQuest API (FastAPI) over HTTPS; the API stores everything in PostgreSQL and runs the AI helpers.

User classes

User	Access	Authenticates with
Parent	Parent area: setup, approvals, rewards, coach, insights	Google sign-in (or the development login); the optional family PIN for approvals
Child	Kid mode only: quests, shop, worlds, check-in	A kid token a grown-up starts, or the family code and their kid PIN on their own device
AI helpers	Run inside the API on a snapshot with children's names replaced by labels	Server-side API key (Groq)

Operating environment

- Modern browsers (Chrome, Edge, Safari, Firefox) on phones, tablets and laptops; layouts from 360 px wide.
- Python 3.12 (FastAPI) and PostgreSQL 17 for the API; any OpenAI-compatible chat API over HTTPS (Groq by default).

Constraints

- Children are 3 to 13; kid mode must work without reading.
- No AI response is applied to family data until a grown-up accepts it (with the PIN, if set).
- All family data lives in PostgreSQL with row-level security; the browser keeps only its sign-in and which celebrations it has shown.
- Groq free tier: about 8,000 tokens a minute per model, about two coach turns a minute.

Assumptions: a family shares a device or gives a child their own; families who share one set a PIN and keep it private.

3. Functional requirements

Each requirement is "must" unless marked otherwise; the acceptance criterion is how a tester confirms it.

ID	Module	Requirement	Acceptance criterion
FR-01	Accounts	Sign in with Google, then create a family (name, time zone, optional 4-digit PIN, your name and role) or join one with an invite code; add a first child (age 3-13, buddy, routine times)	New account lands on first-time setup; age list offers 3 to 13; a PIN that is not 4 digits is refused
FR-02	Accounts	Sign in and out; several grown-ups share a family; a child signs in on their own device with the family code and kid PIN	Wrong password shows the server's message; a wrong kid PIN is refused
FR-03	Routine	Set each child's routine times; each habit takes its time from its part of the day, on chosen days of the week	Today's quests show only on scheduled days, at their times, in the family's time zone
FR-04	Quests	A child taps "I did it!"; status becomes waiting	Card shows "Waiting for a grown-up"
FR-05	Approvals	A grown-up approves or sends back ("not yet"), with the PIN if set	Approve adds Sparks equal to the tier (at most 6 a day) and points, once; a second decision is refused
FR-06	Plan	A parent picks up to 6 habits per child from the catalog, filtered by age group and unlocked tier, or from the family's own custom habits (up to 30)	A locked tier shows how to unlock it; a 7th habit is refused
FR-07	Points	Points = tier base (10, 15, 20) x multiplier; multiplier = 1 + golden + streak (3+ days) + double day + welcome back, max 3	Unit tests reproduce 10 / 20 / 30 point cases

ID	Module	Requirement	Acceptance criterion
FR-08	Levels	Six levels, one per world, at 0, 30, 90, 180, 300 and 450 Sparks; Mosswood Forest (30) unlocks tier 2 and Sky Islands (90) tier 3	Level-up shows a celebration naming the new tier
FR-09	Worlds	6 worlds along one path, with landmarks found as Sparks grow	Locked world shows Sparks needed
FR-10	Shop	A child requests a reward they can afford; points are held	Balance drops by the cost; request shows "Asking a grown-up"
FR-11	Redemption	A grown-up approves (PIN if set) or declines a request	Decline refunds the points; approved screen rewards add minutes to that day's screen quest
FR-12	Mystery box	A random prize from a weighted list; a jackpot pays 100 points at once	The prize shown equals the prize saved
FR-13	Shop admin	A parent adds, prices and removes rewards	Reward names pass the local guardrail
FR-14	Chest	Completing every quest in a day opens one chest: an accessory or 5-25 points	Only once per day
FR-15	Accessories	8 buddy accessories can be collected and worn	Wardrobe shows owned and mystery slots
FR-16	Surprises	A parent plants a note (0-100 bonus points) or a double-points day for one or all children	Shown once as a gift; points credited once
FR-17	Check-in	A child sends mood, trickiest and favourite quest, stickers, and a note (age 6+)	One check-in per child per day; parent sees it in "From your kids"
FR-18	Time limit	A parent sets a daily kid-mode limit (off, 15-60 min)	Warning at 2 min left; break screen at 0; a grown-up can add minutes (with the PIN, if set)
FR-19	Kid mode lock	Leaving kid mode or opening a parent page needs the PIN when the family has one	Parent pages show the PIN dialog while kid mode is on

ID	Module	Requirement	Acceptance criterion
FR-20	Dashboard	Show approvals, reward requests, child stats (level, points, streak, app minutes), check-ins, surprises, AI helpers	Approvals, the PIN and error messages are covered by screen tests

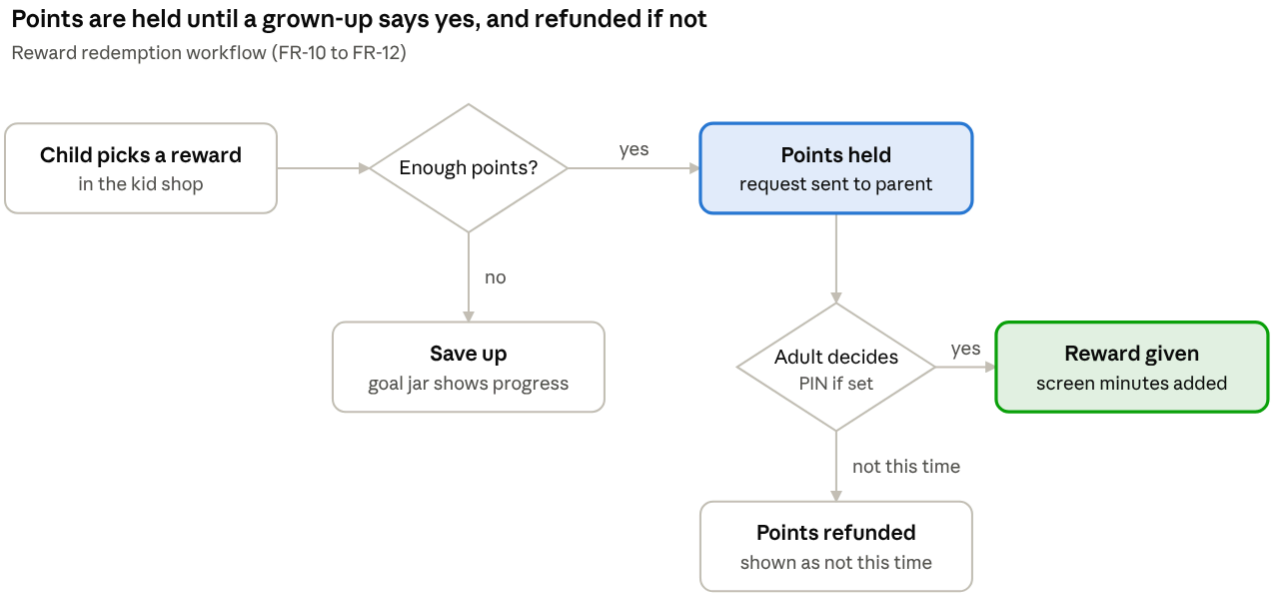
Workflow: quest lifecycle



quest lifecycle · 5 states

Only a grown-up (with the PIN, if set) can move a quest out of "Waiting"; a second decision is refused, so a quest pays out exactly once.

Workflow: reward redemption

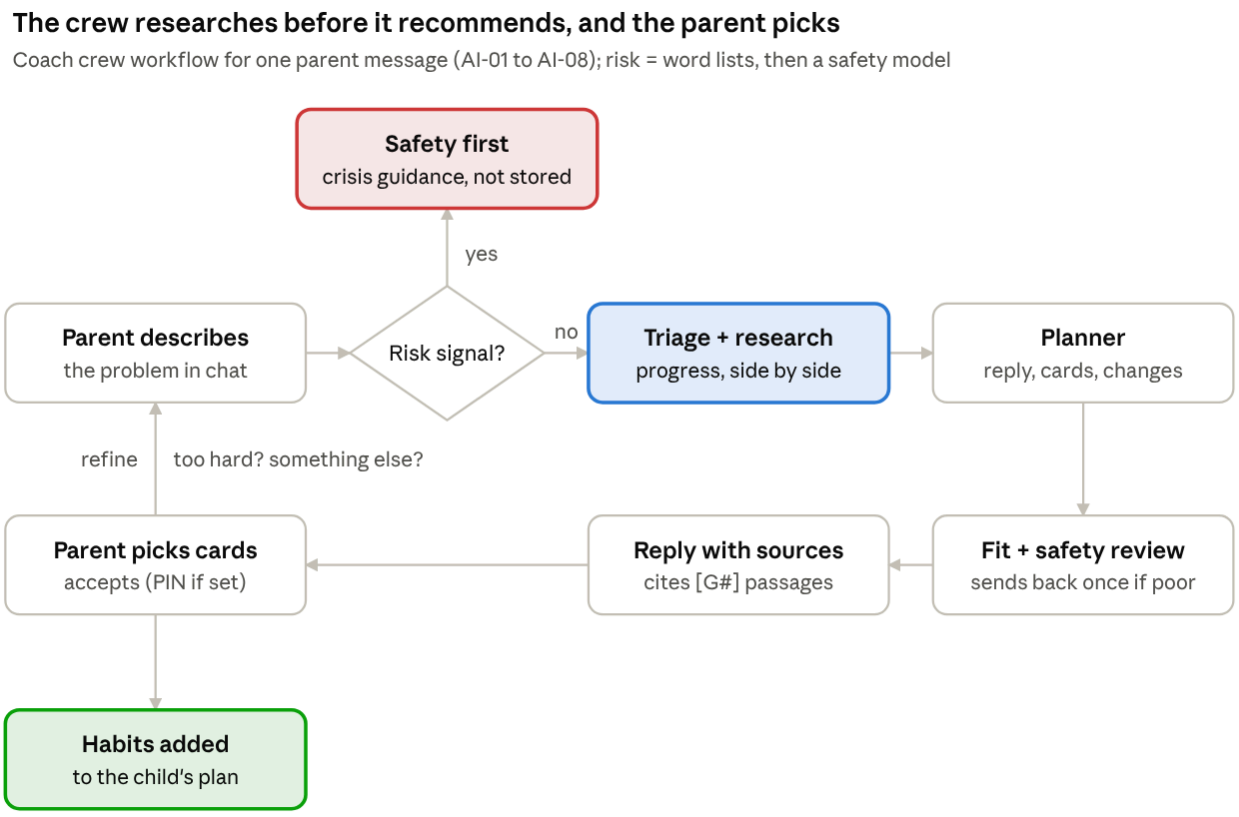


reward redemption · 2 decisions, 3 outcomes

A mystery box reveals its prize at the "Points held" step; a jackpot (100 points) pays out at once and skips the parent decision.

4. AI agent requirements

All agents run inside the API through an OpenAI-compatible provider (Groq by default). The coach is a crew of small agents: triage, research and progress on `openai/gpt-oss-20b`, the planner and fit check on `openai/gpt-oss-120b`, and a safety reviewer on `gpt-oss-safeguard-20b`. The quest planner and weekly review use `gpt-oss-120b`, buddy lines `gpt-oss-20b`. When a model fails, the coach falls back to keyword matching.



coach crew workflow · risk check, crew steps, refine loop

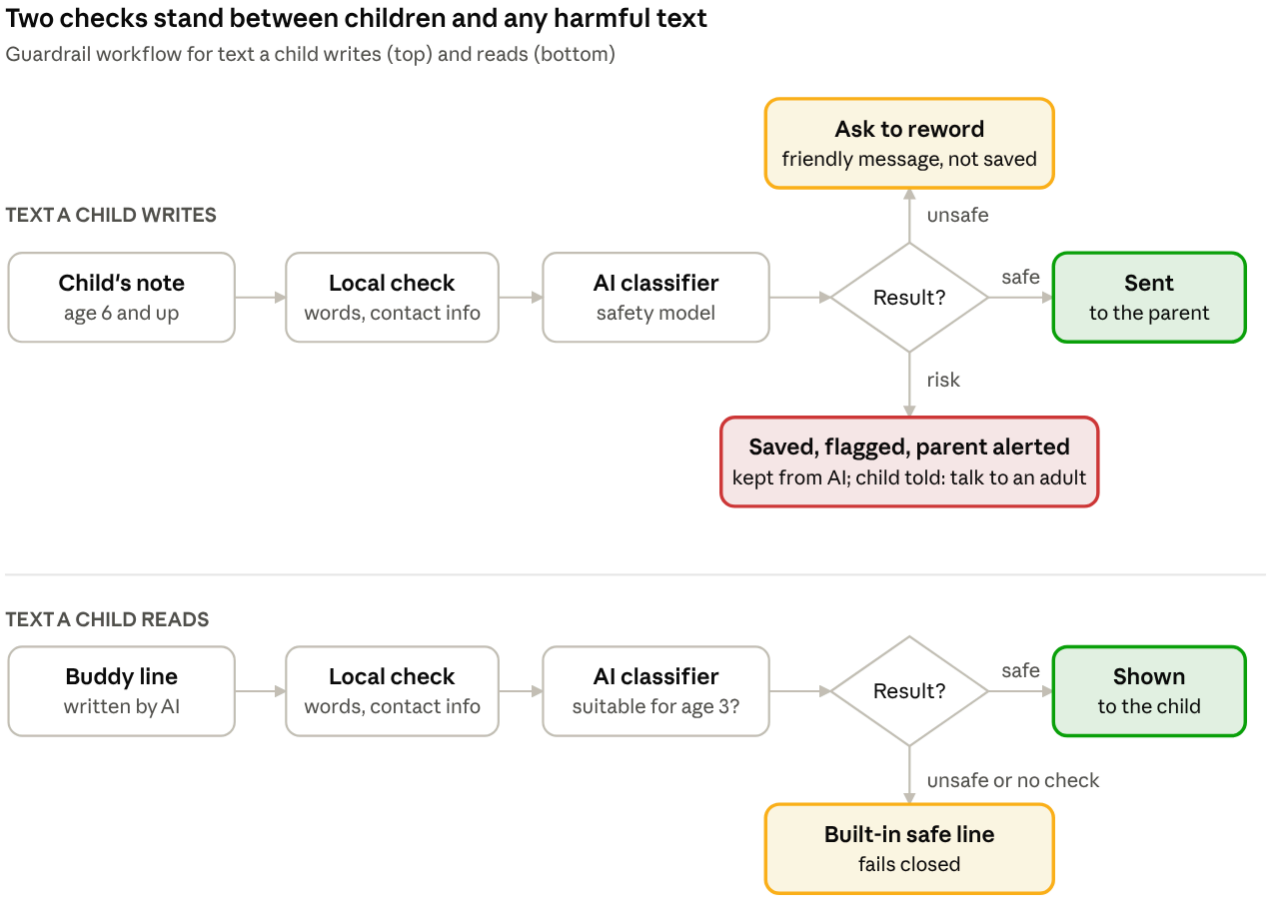
The coach never changes the family itself: its cards wait for a grown-up to accept them, and "too hard" or "something different" sends the parent round the loop again.

ID	Agent	Requirement	Acceptance criterion
AI-01	Coach	Accept a free-text problem and an optional "talking about" child	Typical turn answers within 30 s
AI-02	Coach	Ask one clarifying question when the child or problem is unclear	Vague input gets a question
AI-03	Coach (RAG)	Triage writes up to 3 searches; research ranks guidance passages with BM25 and keeps up to 6, filtered by age; only habits that fit the child's age and unlocked tier can be suggested	Unit tests on retrieval and age filter
AI-04	Coach (RAG)	Cite guidance inline as [G#]; the UI shows cited titles	Sources line under the reply
AI-05	Coach	Recommend 1-3 available quests per child as cards	Locked, wrong-age or unknown quests refused server-side

ID	Agent	Requirement	Acceptance criterion
AI-06	Coach	Refine on request; a new set replaces the old one for that child	Cards change after a refine message
AI-07	Coach	Propose routine changes as cards; never claim a change is made	Applies only after a grown-up accepts (PIN if set)
AI-08	Coach	Know approvals and dismissals in the next turn	Next message carries an [App] note
AI-09	Planner	Up to 4 routine changes and 3 tips as validated JSON	Invalid times or habits dropped
AI-10	Reviewer	Headline, per-child win and watch, up to 3 patterns, up to 2 changes	Uses 14 days and check-ins
AI-11	Buddy	One line of 18 words or fewer for greet, submitted, celebrate	Cached daily; parent can turn it off
AI-12	All	No agent can change Sparks, points, approvals or redemptions	No tool exists for these
AI-13	All	When a model call fails or times out, degrade step by step: research passes passages on, the planner falls back to keyword matching, the safety reviewer fails closed	The parent always gets an answer or a clear message

5. Safety and guardrail requirements

HabbitQuest serves children aged 3 to 13, so every path where a child reads or writes text passes a local check and an AI check.



guardrail workflow · 2 lanes, 2 checks each

Written notes fail open to the local check (a child can always reach a parent); text shown to a child fails closed (a built-in line replaces anything unchecked).

ID	Requirement	Acceptance criterion
SR-01	Block personal info (phone, email, address, links) and explicit, violent, unkind and unsafe content in kid notes	Test phrases blocked with a friendly message
SR-02	Let ordinary feelings through ("I hate homework", "we shot hoops")	No false blocks on the test list
SR-03	Notes that pass locally go to the AI classifier; if it fails, the local result stands	Works without an API key
SR-04	Signs of self-harm or abuse are saved, flagged, shown to the parent as an alert, kept from the AI, and answered with a trusted-adult message	Dashboard shows "Please check in with your child"

ID	Requirement	Acceptance criterion
SR-05	Every AI buddy line passes both checks; otherwise a built-in line is shown	Fails closed
SR-06	Coach recommendations must be library quests that fit the child's age and unlocked tier	Server refuses; browser re-checks
SR-07	A parent message suggesting a child at risk (word lists, then a safety model reading the last three messages) gets emergency and helpline guidance instead of coaching, with a UI banner; nothing is stored or sent on	"wants to die" shows the banner
SR-08	The coach never suggests physical punishment, shaming, threats, withholding food or sleep, or removing earned rewards	Red-team review before launch
SR-09	Parent surprise messages and reward names pass the local check	Blocked text shows an error
SR-10	Agents never receive emails, PINs or children's names (labels instead); custom habits are named only by their goal	Backend tests on the snapshot and the crew's prompts
SR-11	Children under 6 use stickers only	Text box hidden under 6
SR-12	A safety reviewer checks every coach reply; if it fails or finds a problem twice, the unchecked reply is never shown	Fails closed (test_crew.py)

6. Non-functional requirements

The game must feel instant; only AI features may take seconds, and the game keeps working when AI is down.

ID	Quality	Requirement	Measure
NFR-01	Performance	Quest, approval and shop actions respond instantly	Under 1 s on a warm server; a sleeping free server can take a minute, with a "waking up" notice

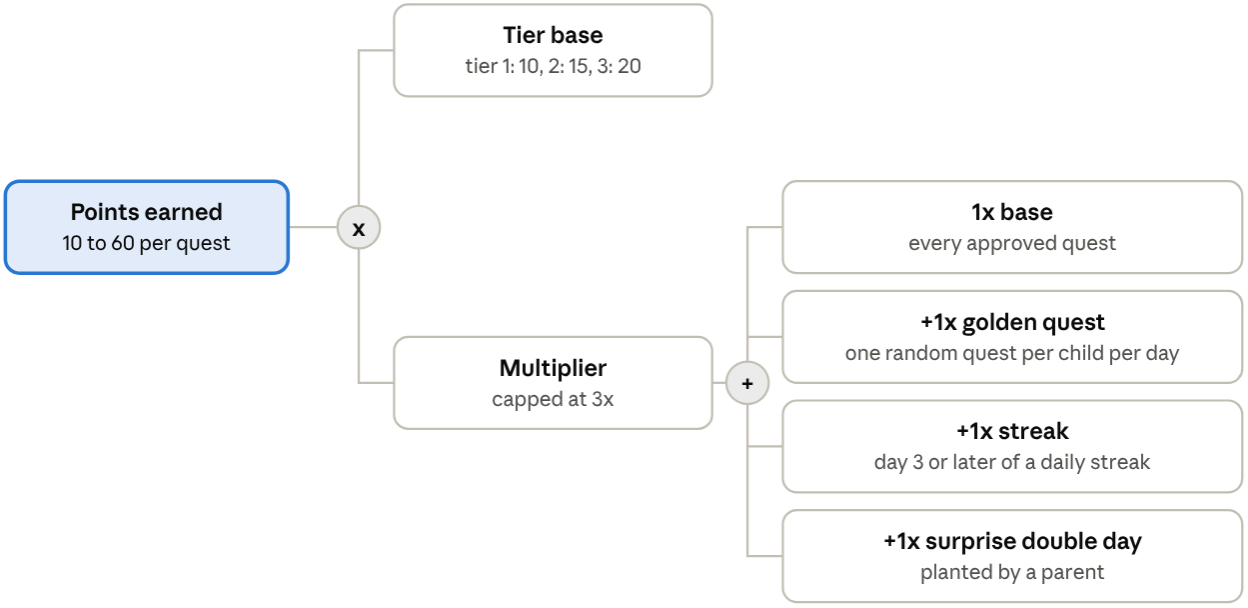
ID	Quality	Requirement	Measure
NFR-02	Performance	Coach turn completes	95% under 30 s; typical 3-10 s
NFR-03	Performance	Buddy line appears	Under 5 s; cached for the day
NFR-04	Availability	The game works when the AI provider is down	AI features fall back to keywords or built-in lines
NFR-05	Reliability	Rate limits handled without errors where possible	Reads retry while the server wakes; a write is never sent twice; AI falls back to keywords
NFR-06	Security	API key never reaches the browser	Bundle scan finds no key or SDK
NFR-07	Security	The family PIN, when set, gates approvals, redemptions, settings changes, accepting AI suggestions and leaving kid mode; the server checks it	PIN valid for 5 minutes after entry
NFR-08	Privacy	Minimum data to AI: labels instead of names, ages, quests, statuses, safe check-ins; row-level security keeps each family's data apart	Backend tests on the snapshot and on row-level security
NFR-09	Usability	Kid mode usable by non-readers	Pictures, clocks, emoji on every quest
NFR-10	Accessibility	Keyboard reachable, labelled controls, reduced-motion support	Manual audit to WCAG 2.2 AA
NFR-11	Responsive	Layout works from 360 px to desktop	No horizontal scroll
NFR-12	Maintainability	Every module is tested; backend coverage stays at 100%	383 backend, 67 frontend and 2 browser tests pass in CI
NFR-13	Cost	AI cost per active family	Under $0.20 a month (target)

7. Data requirements and business rules

The economy has two currencies on purpose: Sparks only go up and drive progress; points go up and down and drive rewards.

A quest pays its tier base times a 1x to 3x multiplier

Points rule applied when a parent approves a quest (FR-07)

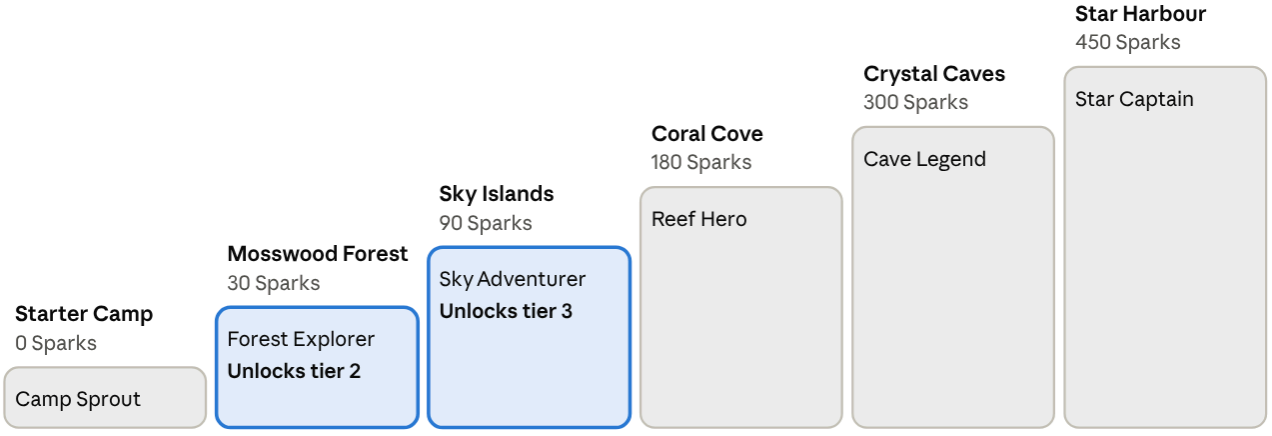


points rule · base times multiplier

A tier 3 golden quest on day 3 of a streak pays the maximum: 20 x 3 = 60 points.

Mosswood Forest and Sky Islands unlock harder quest tiers

Six worlds, each one a level, by lifetime Sparks; not to scale (FR-08, FR-09)



A quest earns Sparks equal to its tier, at most 6 a day; landmarks along the path are found between worlds

level ladder · 6 worlds, 2 tier unlocks

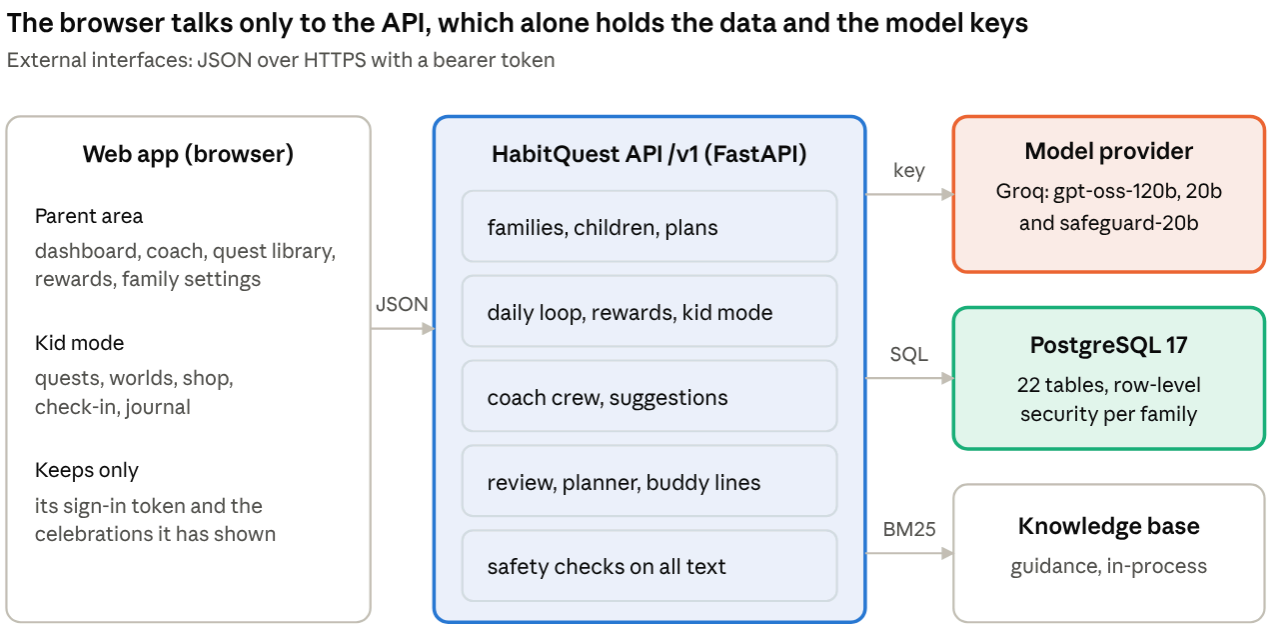
Levels never go down, because Sparks are never taken away.

Core data entities

Area	Tables (schema hq)	Rules
People	users, families, caregivers, children	A Google account links a user; a family has one or more caregivers and an optional PIN hash; children are 3-13, with an optional kid PIN
Catalog and plan	goals, habits, pain_points, child_goals, child_habits	Catalog habits are read-only; a family's custom habits (ids starting "my-") are seen only by that family; a plan holds up to 6 active habits, and removed ones stay as history
Daily loop	completions, points_ledger	One completion per habit per day; a kid can only submit today's; the ledger is append-only and a trigger keeps every balance at 0 or above
Rewards	rewards, reward_requests	A request holds its points until approved, and refunds them when declined
Kid extras	child_extras, checkins, surprises, surprise_reveals, family_settings	One check-in per child per day (a newer one replaces it); each surprise opens once per child
AI	coach_messages, ai_suggestions, ai_usage, ai_reports	Coach messages kept 90 days; a suggestion applies only when a grown-up accepts it; a daily AI limit per family

Other rules: every family table has row-level security (backend/docs/schema.md has the diagrams); kid mode reports time on screen to the server against a daily limit of off, 15, 20, 30, 45 or 60 minutes; the treasure chest opens once a day after every quest is approved (45% chance of a new accessory, otherwise 5-25 points); a mystery box has a parent-set price and a weighted prize list with a rare 100-point jackpot that pays at once.

8. External interfaces



interface map · browser, API, models, database, knowledge base

The API lives under `/v1`, takes and returns JSON, and authenticates with a bearer token (a caregiver's or a kid's). Errors return `{"detail": "<message for people>"}`: 401 sign in again, 403 wrong PIN, 404 missing or another family's, 409 conflict, 422 invalid input, 429 rate limit. The full contract is at `/docs` on the running API and in `backend/docs/frontend-integration.md`.

Area	Main endpoints (under /v1)	Token
Sign-in	POST auth/google, auth/token (development login), kid-login/{code}	None
Families	me, families, families/join, families/{id} with invite-code, login-code, pin, pin/verify, export, delete, caregivers/me	Caregiver
Children and plans	families/{id}/children, children/{id}, children/{id}/plan, child-habits/{id}, catalog, goals, families/{id}/habits, habits/{id}	Caregiver
Daily loop	children/{id}/today, families/{id}/approvals, completions/{id}/approve and /reject	Caregiver

Area	Main endpoints (under /v1)	Token
Rewards	families/{id}/rewards, rewards/{id}, families/{id}/reward-requests, reward-requests/{id}/approve and /decline, children/{id}/redemptions, children/{id}/points	Caregiver
Kid mode	kid/today, kid/completions, kid/journey, kid/extras, kid/chest, kid/app-time, kid/checkins, kid/rewards, kid/reward-requests, kid/surprises, kid/buddy-line	Kid
AI	children/{id}/coach/messages, families/{id}/suggestions, suggestions/{id}/accept and /dismiss, families/{id}/plan-suggestions, reviews, reports	Caregiver

Model interface: any OpenAI-compatible chat completions API (Groq by default, Nebius as a preset), with a JSON schema the provider enforces on every structured answer. The keys stay on the server.

9. Traceability and verification

Every business requirement traces to at least one testable requirement; automated tests cover the rules, and manual checks cover the UI.

Business requirement	Software requirements	Verified by
BR-01 Accounts	FR-01, FR-02, NFR-07	test_accounts.py, test_dev_login.py; sign-in and kid mode screen tests
BR-02 Routine	FR-03	test_onboarding.py, schedule.test.ts
BR-03 Quests	FR-04, FR-05, FR-19, FR-20	test_daily.py; approvals and kid mode screen tests; browser test "a new family's first day"
BR-04 AI coach	AI-01, AI-02, AI-05 to AI-08	test_coach.py, test_crew.py; scripts/eval_coach.py with real models
BR-05 RAG	AI-03, AI-04	test_guidance.py, test_crew.py
BR-06 Safety	SR-01 to SR-12, AI-12	test_safety.py, test_crew.py, test_coach.py
BR-07 Screen time	FR-18	test_family_features.py

Business requirement	Software requirements	Verified by
BR-08 Rewards	FR-07	test_daily.py, test_family_features.py (boosts)
BR-09 Kid feedback	FR-17, SR-04, SR-11	test_family_features.py (check-ins)
BR-10 Levels	FR-08, FR-06	test_journey.py, test_onboarding.py, test_habits.py; Quest library screen tests
BR-11 Redemption	FR-10 to FR-13	test_points.py, test_family_features.py
BR-12 Surprise	FR-12, FR-14 to FR-16	test_family_features.py
BR-13 Insights	AI-09, AI-10	test_insights.py
BR-14 Buddy	AI-11, SR-05	test_insights.py, test_safety.py
BR-15 Worlds	FR-09	test_journey.py, journeyPath.test.ts

In `backend/`, `uv run python -m pytest` runs 383 tests at 100% coverage; in `frontend/`, `npm test` runs 67 logic and screen tests and `npm run e2e` the 2 browser tests. CI runs all three, plus lint, type checks and the build, on every change.